THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 3: **Number Play** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 5	Repeat the largest-minus-smallest step.
2	**(b)** 24	List the sequence.
3	**(a)** 10	The second digit can be any of 10 digits.
4	**(b)** 165	Add the numbers.
5	**(c)** 24	Subtract.
6	**(a)** 4	Find the palindrome.
7	**(b)** 968	Reverse and add.
8	**(a)** 3	Continue the process.
9	**(a)** Both A and R are true and R is the correct explanation of A.	7, 5, 3, 0 in descending order gives 7530.
10	**(a)** Both A and R are true and R is the correct explanation of A.	7 is odd: 3 x 7 + 1 = 22.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Largest minus smallest each time; $1234 \to 3087 \to 8352 \to 6174$ (1); Steps $= 3$ (1).	1+1
12	Apply the rule (1); Terms: 15, maximum: 52 (1).	1+1
13	Digit sum $= 12 \times 5 = 60$ (1); Divisible by 3: yes; divisible by 9: no (1).	1+1
14	Largest minus smallest each time; $2002 \to 2178 \to 7443 \to 3996 \to 6264 \to 4176 \to 6174$ (1); Steps $= 6$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Number $1551 = 1551$; $1551 \div 11 = 141$ exactly (1); Digit sum $= 12$ (1); Palindrome reads the same backwards (1).	1+1+1
16	Build the row; Row: 2, 3, 5, 8, 13, 21, 34, 55 (1); 8th number $= 55$ (1); Sum $= 141$ (1).	1+1+1
17	Arrangements: $4 \times 3 \times 2 \times 1 = 24$ (1); Largest 9752, smallest 2579 (1); Difference 7173 (1).	1+1+1
	OR: Number $2882 = 2882$; $2882 \div 11 = 262$ exactly (1); Digit sum $= 20$ (1); Palindrome reads the same backwards (1).	1+1+1
18	Build the row; Row: 1, 4, 5, 9, 14, 23, 37, 60 (1); 8th number $= 60$ (1); Sum $= 153$ (1).	1+1+1
	OR: Arrangements: $4 \times 3 \times 2 \times 1 = 24$ (1); Largest 8631, smallest 1368 (1); Difference 7263 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) 8530 (1). (ii) 3058 (1). (iii) $8530 - 3058 = 5472$ (1). (iv) An odd number must end in 5 or 3. The smallest start is 3 then 0, giving 3058 (even) or 3085; 3085 ends in 5 and is the smallest odd number (2).	1+1+1+2
20	Number 24542 (1); Digit sum 17 (1); Not divisible by 9 (1); Smallest number to add: 1 (1); Check: new digit sum is a multiple of 9 (1).	1+1+1+1+1
	OR: Reverse-and-add repeatedly (1); Sums: 143, 484 (1); Steps $= 2$ (1); Palindrome 484 has 3 digits (odd) (2).	1+1+1+2

Mistakes to avoid:

- Writing 0 first when forming the smallest number (e.g. 046); the smallest 3-digit number from 4, 0, 6 is 406.
- In a Kaprekar step, subtracting the larger from the smaller or adding the two arrangements.
- Applying the odd rule ($3n + 1$) to an even number or halving an odd number in the hailstone game.
- Reversing digits wrongly in reverse-and-add (the reverse of 68 is 86) or stopping one step early.
- Counting the starting number as a step when counting steps to reach 1 or a palindrome.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 3: **Number Play** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 10	The second digit can be any of 10 digits.
2	**(a)** 228	Add the numbers.
3	**(a)** 24	Subtract.
4	**(b)** 3	Find the palindrome.
5	**(b)** 827	Reverse and add.
6	**(a)** 3	Continue the process.
7	**(a)** 6	Repeat the largest-minus-smallest step.
8	**(b)** 18	List the sequence.
9	**(c)** A is true but R is false.	A is true (5432 - 2345 = 3087). A Kaprekar step subtracts the smallest from the largest arrangement, so R is false.
10	**(d)** A is false but R is true.	046 is not a 3-digit number; the smallest is 406. So A is false and R is true.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Apply the rule (1); Terms: 21, maximum: 52 (1).	1+1
12	Digit sum $= 14 \times 7 = 98$ (1); Divisible by 3: no; divisible by 9: no (1).	1+1
13	Largest minus smallest each time; $4651 \to 5085 \to 7992 \to 7173 \to 6354 \to 3087 \to 8352 \to 6174$ (1); Steps $= 7$ (1).	1+1
14	Apply the rule (1); Terms: 11, maximum: 40 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Number 3663 $= 3663$; $3663 \div 11 = 333$ exactly (1); Digit sum $= 18$ (1); Palindrome reads the same backwards (1).	1+1+1
16	Build the row; Row: 3, 5, 8, 13, 21, 34, 55 (1); 7th number $= 55$ (1); Sum $= 139$ (1).	1+1+1
17	Arrangements: $4 \times 3 \times 2 \times 1 = 24$ (1); Largest 9654, smallest 4569 (1); Difference 5085 (1).	1+1+1
	OR: Number 4994 $= 4994$; $4994 \div 11 = 454$ exactly (1); Digit sum $= 26$ (1); Palindrome reads the same backwards (1).	1+1+1
18	Build the row; Row: 2, 6, 8, 14, 22, 36, 58, 94 (1); 8th number $= 94$ (1); Sum $= 240$ (1).	1+1+1
	OR: Arrangements: $4 \times 3 \times 2 \times 1 = 24$ (1); Largest 9821, smallest 1289 (1); Difference 8532 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) 3 (1). (ii) $3 \times 3 + 1 = 10$ (1). (iii) 6, 3, 10, 5, 16, 8, 4, 2, 1 is 8 steps (1). (iv) 5, 16, 8, 4, 2, 1 (1); 5 steps and the largest number is 16 (1).	1+1+1+2
20	Number 46264 (1); Digit sum 22 (1); Not divisible by 9 (1); Smallest number to add: 5 (1); Check: new digit sum is a multiple of 9 (1).	1+1+1+1+1
	OR: Reverse-and-add repeatedly (1); Sums: 143, 484 (1); Steps $= 2$ (1); Palindrome 484 has 3 digits (odd) (2).	1+1+1+2

Mistakes to avoid:

- Writing 0 first when forming the smallest number (e.g. 046); the smallest 3-digit number from 4, 0, 6 is 406.
- In a Kaprekar step, subtracting the larger from the smaller or adding the two arrangements.
- Applying the odd rule ($3n + 1$) to an even number or halving an odd number in the hailstone game.
- Reversing digits wrongly in reverse-and-add (the reverse of 68 is 86) or stopping one step early.
- Counting the starting number as a step when counting steps to reach 1 or a palindrome.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 3: **Number Play** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 26	Subtract.
2	**(c)** 2	Find the palindrome.
3	**(a)** 1292	Reverse and add.
4	**(a)** 6	Continue the process.
5	**(c)** 3	Repeat the largest-minus-smallest step.
6	**(a)** 16	List the sequence.
7	**(a)** 510	Add the numbers.
8	**(d)** 18	Subtract.
9	**(b)** Both A and R are true but R is not the correct explanation of A.	1 + 2 + 3 = 6 and 123 = 3 x 41. Both are true, but R does not explain why the digit sum is 6.
10	**(a)** Both A and R are true and R is the correct explanation of A.	Middle row: 3+5 = 8 and 5+2 = 7; top: 8+7 = 15.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Digit sum $= 11 \times 4 = 44$ (1); Divisible by 3: no; divisible by 9: no (1).	1+1
12	Largest minus smallest each time; $8532 \rightarrow 6174$ (1); Steps $= 1$ (1).	1+1
13	Apply the rule (1); Terms: 19, maximum: 160 (1).	1+1
14	Digit sum $= 15 \times 6 = 90$ (1); Divisible by 3: yes; divisible by 9: yes (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Number 5225 $= 5225$; $5225 \div 11 = 475$ exactly (1); Digit sum $= 14$ (1); Palindrome reads the same backwards (1).	1+1+1
16	Build the row; Row: 4, 7, 11, 18, 29, 47, 76 (1); 7th number $= 76$ (1); Sum $= 192$ (1).	1+1+1
17	Arrangements: $4 \times 3 \times 2 \times 1 = 24$ (1); Largest 9743, smallest 3479 (1); Difference 6264 (1).	1+1+1
	OR: Number 7447 $= 7447$; $7447 \div 11 = 677$ exactly (1); Digit sum $= 22$ (1); Palindrome reads the same backwards (1).	1+1+1
18	Build the row; Row: 1, 5, 6, 11, 17, 28, 45, 73, 118 (1); 9th number $= 118$ (1); Sum $= 304$ (1).	1+1+1
	OR: Arrangements: $4 \times 3 \times 2 \times 1 = 24$ (1); Largest 8762, smallest 2678 (1); Difference 6084 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) 75 (1). (ii) 132 (1). (iii) $132 + 231 = 363$ is a palindrome, so 2 steps (1). (iv) $68 + 86 = 154$, $154 + 451 = 605$, $605 + 506 = 1111$ (1); 3 steps and the palindrome is 1111 (1).	1+1+1+2
20	Number 63936 (1); Digit sum 27 (1); Divisible by 9 (1); Smallest number to add: 0 (1); Check: new digit sum is a multiple of 9 (1).	1+1+1+1+1
	OR: Reverse-and-add repeatedly (1); Sums: 483, 867, 1635, 6996 (1); Steps $= 4$ (1); Palindrome 6996 has 4 digits (even) (2).	1+1+1+2

Mistakes to avoid:

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